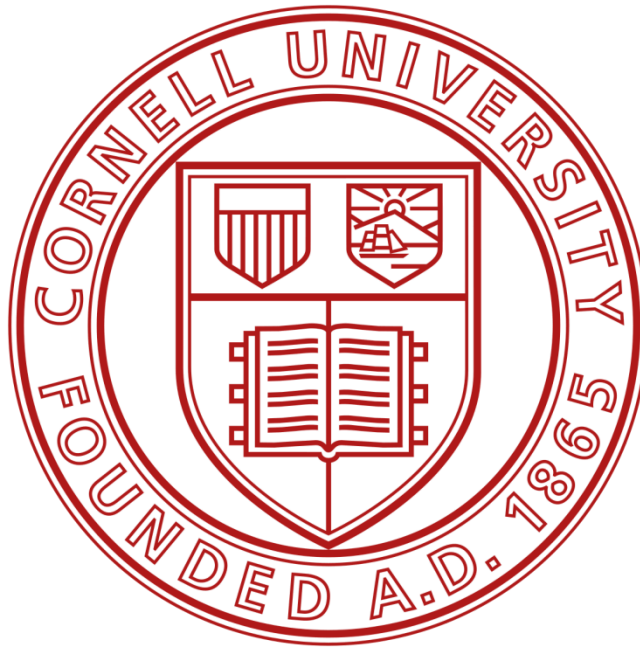




Clinical Preceptorship Final Report

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Introduction

Over the Fall 2025 semester, I completed seven clinical immersions (15-20 hours) at Guthrie Robert Packer Hospital under the mentorship of Dr. Vineet Agrawal (urology) and surgical preceptors Dr. Firdos Ziauddin (breast surgery) and Dr. Chalairat Suk-Ouichai (urology). Across breast excisional biopsies and partial mastectomies, blue-light cystoscopy-guided bladder tumor resections, ureteroscopic laser lithotripsy with stent placement, and outpatient urology clinics, I observed how biomedical instrumentation, workflow, and human factors converge in real clinical environments.

This report synthesizes those experiences through a biomedical engineering lens, with particular attention to signal acquisition and processing, energy-delivery devices, imaging systems, anesthesia monitoring, and information-management technologies. Alongside recounting each immersion, I learned more about how devices are integrated into surgical workflow, how design decisions influence safety and usability, and how technology supports - but never replaces - clinical judgment. These encounters substantially deepened my understanding of how theoretical engineering concepts translate into patient-centered care.

1. Breast Surgery and Radar-Guided Localization

1.1 Right Breast Excisional Biopsy: Establishing a Baseline

My first immersion, a right breast excisional biopsy with Dr. Ziauddin, provided a foundation for understanding the operating room as a tightly coordinated system. Standard instruments such as scalpels, Metzenbaum scissors, Adson and Lahey forceps, hemostats, retractors, and nylon and silk sutures were complemented by a suite of biomedical devices including the Skytron 6702 Hercules surgical table, an electrosurgical unit with monopolar grounding pads, an anesthesia machine, a smoke evacuation system, and a Bair Hugger warming system. Together, these tools maintained a safe and controlled environment in which the surgeon could focus on tissue handling and decision-making.

The procedure also highlighted how process design extends beyond the immediate visual field. A color-coded tissue staining protocol was used to mark superior, inferior, medial, lateral, anterior, and posterior aspects of the specimen before it was placed in formalin and sent to pathology. This simple but carefully designed system ensures that pathologists can correctly orient the mass and correlate microscopic findings with anatomic location, illustrating how engineering principles of standardization and error prevention are embedded in routine clinical practice.

Minor disruptions during the case, such as an electrosurgical alarm due to a loose component and brief patient movement as anesthesia depth was adjusted, underscored the need for responsive, fault-tolerant systems. Design details like cable clutter on the floor and gloves that were slightly loose or wrinkled became surprisingly salient, as they interfered with ergonomics and increased tripping or contamination risk due to their occasional entanglement on instruments. Observing these small but persistent pain points immediately framed the OR as a real-world laboratory for biomedical engineering and suggested opportunities for wireless instrumentation, improved glove materials, and more intuitive equipment layouts.

1.2 Palpation to Radar: SCOUT®-Guided Excision

Later that same day, and again in a subsequent immersion, I observed reflector-localized breast surgeries that used the SCOUT® radar localization system. These cases contrasted sharply with

the palpation-guided biopsy and demonstrated how a single new technology can reshape workflow, ergonomics, and cognitive load.

The SCOUT system relies on a miniature biocompatible reflector implanted in the breast days to weeks prior to surgery, a console that transmits micro-impulse radar and infrared light, and a sterile handpiece that receives reflected pulses and converts them into numerical distance readouts and auditory beeps. In the OR, this translated into a simple but powerful sensory cue: as the surgeon's handpiece approached the target, the "beeps" accelerated and the distance value decreased. The system allowed Dr. Ziauddin to plan incisions based on both oncologic and cosmetic considerations, rather than being constrained by the entry path of a wire placed on the morning of surgery.

From an instrumentation standpoint, SCOUT embodies signals-and-systems concepts. The device must transmit and receive radar pulses through heterogeneous tissue, suppress noise, and map signal characteristics to distance with sub-millimeter precision, all while presenting the result on a minimalist interface that does not compete with other monitors and alarms. Dense breast tissue and small angular deviations of the handpiece occasionally led to subtle artifacts, illustrating design trade-offs among penetration depth, reflector size, and angular sensitivity. Nevertheless, the system consistently acted as a precision amplifier rather than a substitute for the surgeon's expertise; tactile feedback, anatomical knowledge, and visual cues remained central, with SCOUT providing an additional layer of confidence around margin control.

These immersions reinforced the idea that successful biomedical devices are those that sharpen human judgment without overwhelming the user. Interface simplicity became a safety feature in itself: a console with only essential controls and a clear, sterilizable display allowed the surgeon to integrate radar guidance into existing mental models and workflows rather than contending with a steep learning curve during a high-stakes operation.

2. Urologic Surgery: Light, Lasers, and Imaging

2.1 Blue-Light Cystoscopy-Guided TURBT

My first immersion with Dr. Suk involved a transurethral resection of a bladder tumor (TURBT) under blue-light cystoscopy. After intravesical instillation of a photosensitizing agent, malignant cells accumulated protoporphyrin IX and fluoresced bright pink under blue/violet illumination, while the surrounding mucosa remained relatively dark. On the monitor, the tumor resembled sea coral, a description the surgeon used to help us appreciate the subtle topography and color differences.

The case showcased several layers of instrumentation. A camera-equipped cystoscope transmitted real-time video to a large monitor, allowing the whole team to share a magnified view of the bladder. Spectral engineering, carefully chosen excitation wavelengths and optical filters, enabled selective visualization of tumor fluorescence. The Storz cutting system delivered high-frequency electrical current through a loop electrode to shave away tumor tissue while irrigation and suction maintained visibility and removed debris.

Beyond technical performance, the procedure exposed system-level issues. Manual transfer of the patient from a rolling bed to the Skytron table required significant physical effort from multiple staff members and highlighted the potential value of powered lifts or improved transfer aids. Cables from suction, irrigation, and electrosurgical units tangled underfoot, raising both safety and efficiency concerns. A post-case tour of the machine and supply rooms revealed the extensive behind-the-scenes infrastructure required to keep these devices clean, functional, and ready for use, including waste management pathways and maintenance logs.

This immersion made it clear that device design cannot stop at the distal tip of a scope. Engineers must consider how instruments are stored, cleaned, moved, and integrated into cramped, busy environments. A technically elegant system that burdens staff with cumbersome transfers, confusing cable layouts, or complex cleaning requirements is not truly optimized for clinical reality.

2.2 Ureteroscopy, Laser Lithotripsy, and Stent Placement

Another immersion with Dr. Suk focused on ureteroscopic stone management with laser lithotripsy and ureteral stent placement. Here, imaging and energy delivery were even more tightly coupled. A flexible digital ureteroscope was advanced through the urinary tract to visualize the stone, while a Ho:YAG laser fiber was used to fragment it under direct vision. Real-time fluoroscopy from a Philips C-arm provided positional confirmation of the scope, stone, and eventual stent.

Laser lithotripsy offered a vivid demonstration of energy-tissue interaction. Pulses of infrared light at 2100 nm, strongly absorbed by water, produced micro-bubble expansion and photothermal ablation at the stone surface while limiting penetration into surrounding tissue. By adjusting pulse energy, frequency, and duration, the surgical team could toggle between dusting the stone into fine particles and fragmenting it into larger pieces for basket retrieval. Watching the stone gradually crumble without any external incision made abstract concepts like wavelength selection and absorption depth feel tangible.

The C-arm X-Ray machine added another dimension. Wrapped in sterile plastic to preserve the field, it offered limited degrees of freedom, so careful coordination was required to position it without interfering with staff or equipment. A detailed conversation with the radiologic technologist revealed a nuanced view of technological progress. Newer C-arm models offer lower radiation doses and more sensitive detectors, but their touch-screen interfaces and crowded control panels can be difficult to operate with gloved hands. Similarly, laser safety goggles provided adequate optical protection but proved to be uncomfortable enough that staff sometimes hesitated to wear them consistently.

These details underscored the importance of human-factors engineering. Protective equipment must be comfortable to encourage adherence; otherwise, safety gains are theoretical rather than real. Interfaces with excessive features can slow clinical workflow, particularly when sterility and time pressure limit the ability to navigate nested menus. Generational differences in equipment preferences, such as favoring a simple single-dot targeting system over newer dual-beam lasers, also reminded me that device adoption depends on familiarity and trust, not just technical specifications.

Throughout the procedure, Dr. Suk's light-hearted, warm, and welcoming demeanor connected these engineering observations back to patient care. She moved fluidly between imaging interpretation, ureteroscope navigation, and laser activation while maintaining clear communication with the team. Her ability to manage complex instrumentation without appearing rushed highlighted the responsibility engineers have to design tools that streamline, rather than complicate, the surgeon's cognitive workload.

3. Outpatient Informatics

3.1 Outpatient Urology and EMR Integration

Shadowing Dr. Agrawal in the outpatient clinic offered a different perspective on biomedical systems, one focused on information management and patient interaction rather than high-intensity instrumentation. Over a series of rapid consultations addressing kidney stones, prostate surgery follow-up, varicocele, and urinary symptoms, I watched him navigate Epic with remarkable efficiency while maintaining strong rapport with patients. He minimized technology use in the room to stay fully present, then harnessed AI-assisted tools outside the room - voice commands, smart templating, and structured note copying - to document encounters and manage a high patient volume.

Although some aspects of the Epic interface felt dated from a user-experience perspective, its ability to aggregate laboratory results, imaging reports, operative notes, and referral histories clearly supported clinical decision-making. At the same time, episodes of misdirected referrals and minor communication lapses highlighted that software alone cannot guarantee smooth care pathways. Workflow design, training, and institutional culture all influence how effectively such systems are used.

This experience broadened my view of biomedical engineering to include health information technology and workflow optimization. EMR design shapes not only clinician efficiency but also the clarity with which patient stories are captured and communicated across teams. Thoughtful integration of AI tools has the potential to relieve documentation burden, but only if done in a way that preserves the primacy of the clinician-patient interaction.

4. Engineering Themes

4.1 Signals and Systems in the OR

Across breast and urologic procedures, I repeatedly encountered the theme of signals and systems. SCOUT radar localization, blue-light fluorescence, Ho:YAG laser pulses, BIS indices, and even the auditory feedback of suction machines all represent signals that must be acquired, processed, and translated into intuitive information. Clinicians rarely act on raw data alone; they rely on trended displays, multi-modal feedback, and their own experiential models of how physiology and devices behave over time.

These observations made classroom topics such as sampling, filtering, noise suppression, and sensor fusion feel concrete. The reliability of a single number on a screen depends on a chain of design decisions, from transducer physics to analog circuitry and digital algorithms. In the OR, failure at any point in that chain can compromise safety, whereas robust design can empower clinicians to act decisively.

4.2 Anesthesia Machine and BIS Monitoring

During a reflector-localized partial mastectomy and sentinel lymph node biopsy with Dr. Ziauddin, I deliberately shifted my attention from the operative field to the anesthesia workstation. Under the guidance of a generous CRNA, I learned how the machine continuously monitors and records heart rate, non-invasive blood pressure, oxygen saturation, respiratory rate, end-tidal CO₂, temperature, and depth of anesthesia via a Bispectral Index (BIS) monitor. These data streams are automatically captured in the Epic electronic medical record, with regular manual confirmation.

Viewed through an engineering lens, the anesthesia station is a real-time control system. Multiple sensors acquire physiological signals that are filtered, averaged, and trended, enabling the CRNA to adjust gas concentrations and medications to keep the patient within a safe anesthetic window. BIS, although not mandated by professional societies and subject to artifacts, provides an additional trend indicator of sedation depth, complementing more traditional assessments such as movement, hemodynamics, and physical examination.

I truly began to understand how much the effectiveness of this system depends on human-machine interaction. The CRNA emphasized the importance of being able to interpret subtle changes at a glance, which relies on intuitive screen layouts, meaningful alarm thresholds, and tactile familiarity with knobs and switches. Their preventative mindset, preferring to anticipate instability rather than react to crises, depends on reliable signal processing and clear trend visualization. This immersion transformed anesthesia in my mind from a simple sedation technique to a sophisticated application of biomedical signal processing and control theory.

4.3 Overall Systems and Workflow Integration

The immersions also highlighted the importance of thinking at the system level. Tissue staining protocols connected the OR to the pathology lab. Preventive maintenance schedules and clearly marked “clean/unclean” states on suction machines and smoke evacuators ensured long-term reliability. Supply room organization, machine-cleaning pipelines, and EMR documentation pathways all influenced how efficiently procedures began and ended, how results were communicated, and how quickly complications could be addressed.

These experiences reminded me that biomedical engineers must design not only devices but also the workflows and interfaces through which those devices interact with people and institutions. Interoperability, maintainability, and clear communication channels are as crucial as hardware performance.

5. Final Poster

5.1 Poster Development & Content

A culminating component of this preceptorship was the development of our poster, “Enhancing Surgical Care: Imaging & Laser Technologies in Urology - From Light to Laser: How Biomedical Engineering Shapes Modern Urology”, which I created and presented with my fellow classmate, Shreya Dandwate. Synthesizing our observations into a coherent visual narrative forced us to distill complex workflows into key engineering concepts and to highlight the central role of light-tissue interaction, optics, and energy delivery in modern urologic care. Presenting the poster in class and listening to our peers’ projects was unexpectedly enriching. Many teams had observed different specialties, yet we heard recurring themes of signal interpretation, human-factors challenges, and the importance of clear communication. At the same time, there were striking differences in device types, patient populations, and institutional culture. Comparing these overlaps and contrasts helped situate our experience within a broader spectrum of clinical practice and reinforced that biomedical engineering must remain adaptable to diverse contexts.

6. Personal Takeaways

This preceptorship was not only academically enriching - it was, without question, the most meaningful and memorable course I have taken in my entire academic career. BME 5100 offered an experience that felt genuinely rare, shaped by the extraordinary effort, care, and intentionality poured into the program by our professors and by the clinicians who welcomed us into their workspaces with humility and generosity. The willingness of surgeons, CRNAs, technologists, and clinical staff - individuals operating at the highest levels of expertise - to pause, teach, and allow us into environments seldom accessible to students left a lasting impression on me.

As someone who will be entering industry rather than medical school, I recognize how singular this opportunity was. I have always been fascinated by the inner workings of a hospital, but I never imagined I would one day observe live surgeries, much less find the OR to be such a refreshing and unexpectedly calming environment. Watching teams come together with precision, empathy, and technical mastery was an experience I know I will not have again, which makes it all the more meaningful.

Equally impactful were the moments beyond the procedures themselves, such as the early-morning drives with fellow classmates, the respectful yet warm one-on-one interactions with surgeons, and the casual, reflective conversations in class afterward with professors who celebrated our observations as though they were their own. The entire experience fostered a sense of community, curiosity, and shared purpose that is rare in academic settings.

In many ways, this class felt like a dream come true: it blended my longstanding interest in clinical environments with my passion for engineering, offered unforgettable exposure to real surgical practice, and provided mentorship and camaraderie that I will carry with me long after graduation. It reaffirmed why I chose biomedical engineering in the first place and reminded me that behind every device, every workflow, and every engineering problem, there are human stories, human hands, and human lives being touched.

7. Acknowledgements

I am deeply grateful to Dr. Vineet Agrawal for serving as our preceptor, for welcoming us into his clinic, and for connecting us with surgeons whose practices exemplify technologically advanced yet compassionate care. I am equally thankful to Dr. Firdos Ziauddin and Dr. Chalaïrat Suk-Ouichai, whose generosity in the OR - explaining decisions, answering questions, and modeling exemplary leadership - made this learning experience possible. I would also like to thank the Guthrie urology and breast surgery teams, including nurses, CRNAs, technologists, and support staff, for their patience and willingness to share insights about the devices and workflows they rely on every day. Finally, I have the utmost gratitude for the Cornell BME faculty, particularly Professor Suélia Fleury Rosa and Linds Mulholland, for structuring the preceptorship, encouraging reflective practice, and creating a learning environment where engineering students can meaningfully engage with clinical reality.

Appendix

Appendix A. Supplemental Documentation

 Clinical Preceptorship: Shadowing Notes (Transcribed)

 BME 5911 - Final Poster (Shreya Dandwate & Yashvi Gupta).pdf

Appendix B. Immersion Notes

 10-17-25 Shadowing (Dr. Firdos Ziauddin - 1 & 2).pdf

 10-24-25 Shadowing (Dr. Suk & Agrawal - 3 & 4).pdf

 11-07-25 Shadowing (Dr. Firdos Ziauddin - 5) .pdf

 11-14-25 Shadowing (Dr. Suk & Firdos Ziauddin - 6 & 7) .pdf